

# Microscopic study of the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ and ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reactions in a multiconfiguration three-cluster model

P. Descouvemont<sup>1</sup>, D. Baye

*Physique Nucléaire Théorique et Physique Mathématique, Université Libre de Bruxelles, Campus Plaine, C.P. 229, B-1050 Brussels, Belgium*

Received 26 October 1992

(Revised 9 February 1993)

## Abstract

The three-cluster generator coordinate method is applied to the  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$  and  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  capture reactions. The  ${}^8\text{B}$  (or  ${}^8\text{Li}$ ) nucleus is defined by a mixing of  $(\alpha + {}^3\text{He}) + p$  and  $(\alpha + p) + {}^3\text{He}$  (or mirror) configurations. We investigate the sensitivity of different observables with respect to the nucleon–nucleon interaction by considering four different Volkov forces. For all of them, the model fairly reproduces most of the experimentally known spectroscopic properties of the  ${}^8\text{B}$  and  ${}^8\text{Li}$  mirror nuclei. Recent measurements of the quadrupole moments are well explained without a halo structure. The  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$  cross section supports the data of Imhof et al. rather than those of Wiescher et al. At zero energy, the  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  astrophysical  $S$ -factors slightly depend on the nucleon–nucleon interaction. They are consistent with the currently accepted value, but contradict lower estimates.

**Key words:** NUCLEAR REACTIONS  ${}^7\text{Li}(n, \gamma)$ ,  $E(\text{cm}) \leq 0.8$  MeV; calculated  $\sigma(\theta)$  vs  $E$ .  ${}^7\text{Be}(p, \gamma)$ ,  $E(\text{cm}) < 3$  MeV; calculated astrophysical  $S$ -factor vs  $E$ . Three-cluster generator coordinate method. NUCLEAR STRUCTURE  ${}^8\text{Li}$ ,  ${}^8\text{B}$ , calculated  $\mu$ , quadrupole moment, nucleon width of levels,  $B(\lambda)$ . Multi-configuration three-cluster model.

## 1. Introduction

The  $T = 1$  eight-nucleon systems are interesting in many respects. The  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  reaction is a key process in the solar neutrino problem, since  ${}^8\text{B}$  decay provides high-energy neutrinos which can be observed in the  ${}^{37}\text{Cl}$  detector [1]. If the energy dependence of the  $S$ -factor in the astrophysical region (typically  $E \leq 50$  keV) is well understood [2], its overall normalization is still too inaccurate for a definite conclusion about its influence on the solar-neutrino puzzle. From an experimental point of view, the normalization of the  ${}^7\text{Be}(p, \gamma){}^8\text{B}$   $S$ -factor is

<sup>1</sup> Chercheur qualifié FNRS.

deduced from the  ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$  cross section at 0.77 MeV. The rather large uncertainty on this cross section directly influences the  ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$  data [3]. In spite of many theoretical studies [4–12], the low-energy  $S$ -factor remains controversial.

The mirror  ${}^7\text{Li}(\text{n}, \gamma){}^8\text{Li}$  reaction also deserves a special attention. It is a key process in inhomogeneous big-bang models [13] since it bridges the instability gap at  $A = 8$  and is therefore the starting point for heavy-element production. The first experimental data, due to Imhof et al. [14], are rather well explained by our first microscopic approach [9], but they have recently been questioned by Wiescher et al. [15] who find a reduction by a factor of two. However, new measurements by Nagai et al. [16] are very close to the data of Imhof et al. This current controversy certainly deserves new investigations.

The spectroscopy of  ${}^8\text{Li}$  and, to a lesser extent, of  ${}^8\text{B}$  is very interesting. Recent experiments on Coulomb excitation of  ${}^8\text{Li}$  by heavy targets [17,18] provide  $B(\text{E}2)$  values between the ground and first excited states which are unexpectedly large [19]. In addition, two narrow states ( $3^+$  at 2.26 MeV and  $4^+$  at 6.53 MeV) can not be described by  ${}^7\text{Li}(\text{g.s.}) + \text{n}$  configurations [9], but are assumed to have a more exotic structure [20]. Fujiwara and Tang [21] have performed a multichannel microscopic study of the  ${}^8\text{Li}$  nucleus, but without spin–orbit and Coulomb interactions. They show the importance of the  ${}^7\text{Li}^* + \text{n}$  and  ${}^5\text{He} + \text{t}$  configurations, and the minor effect of the  ${}^4\text{He} + {}^4\text{H}$  configuration.

In 1988, we have applied a microscopic three-cluster model to the  ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$  reaction [9]. The fully antisymmetric wave functions are defined in the generator coordinate method (GCM) involving  $\alpha$  and  ${}^3\text{He}$  clusters, and a proton. Describing the  ${}^7\text{Be}$  (or  ${}^7\text{Li}$ ) nucleus by an  $\alpha + {}^3\text{He}$  (or  ${}^3\text{H}$ ) structure with a relative angular momentum  $L = 1$  provides a rather good description of many  ${}^8\text{B}$  (or  ${}^8\text{Li}$ ) spectroscopic properties. The interest of a simultaneous description of the mirror  ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$  and  ${}^7\text{Li}(\text{n}, \gamma){}^8\text{Li}$  reactions is emphasized in ref. [9]. Similar microscopic studies have been carried out by Kolbe et al. [10] and by Johnson et al. [12].

In the present paper, we reconsider these problems by improving our first GCM three-cluster model [9]. The configuration space is enlarged in two ways; we extend to  $L = 3$  the  $\alpha + {}^3\text{He}$  states which provide the  ${}^7\text{Be}(\frac{7}{2}^-, \frac{5}{2}^-) + \text{p}$  channels, and we add the  ${}^5\text{Li}(\frac{3}{2}^-, \frac{1}{2}^-) + {}^3\text{He}$  rearrangement configurations. We also try to meet an interesting point raised by Mukhamedzhanov and Timofeyuk [11]. These authors stress that the  ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$  capture process mainly depends on the amplitude of the bound-state wave function at large distances. They argue that this amplitude is strongly dependent on the choice of the nucleon–nucleon interaction, and is overestimated by the Volkov [22] interaction V2 which we employ in ref. [9]. Although their conclusion is based on a calculation where the shell-model wave functions are not consistent with the nucleon–nucleon interaction, their criticism deserves further attention [23]. We therefore consider here four different nucleon–nucleon interactions (Volkov forces V1, V2, V3 and V4); this allows us to

investigate the sensitivity of the different physical quantities with respect to the nucleon–nucleon force.

In sect. 2, we briefly present the microscopic three-cluster model. The  $^8\text{Li}$  spectroscopy and the  $^7\text{Li}(n, \gamma)^8\text{Li}$  reaction are analyzed in sect. 3, whereas the mirror systems are investigated in sect. 4. Concluding remarks are given in sect. 5.

## 2. Summary of the model

The microscopic three-cluster model used here is detailed in refs. [9,24,25]. Let us define Slater determinants  $\Phi$  with alpha,  $^3\text{He}$  and proton clusters described in the harmonic-oscillator model (with a common oscillator parameter  $b = 1.50$  fm) and located at three different points. We choose here two configurations (see Fig. 1) labeled by index  $\alpha$ :  $^7\text{Be} + p$  ( $\alpha = 1$ ) where  $^7\text{Be}$  is defined by an  $\alpha + ^3\text{He}$  structure and  $^5\text{Li} + ^3\text{He}$  ( $\alpha = 2$ ) where  $^5\text{Li}$  is defined by an  $\alpha + p$  structure. The latter configuration is not required for studying the  $^7\text{Be}(p, \gamma)^8\text{B}$  capture, but enlarges the configuration space and therefore improves the wave functions. In each case, two generator coordinates  $R_1$  and  $R_2$  are defined according to Fig. 1.

After projection on total angular momentum  $J$  and parity  $\pi$ , the GCM three-cluster wave functions of  $^8\text{B}$  read

$$\Psi^{JM\pi} = \sum_{\text{all}} \sum_{R_1 R_2} f_{\text{all}}^{J\pi}(R_1, R_2) \Phi_{\text{all}}^{JM\pi}(R_1, R_2), \quad (1)$$

where  $l$  is the relative angular momentum between  $^7\text{Be}$  and proton and  $f_{\text{all}}^{J\pi}$  is a generator function. In Eq. (1), the projection also ensures a correct orbital momentum  $L$  for the relative motion of the  $\alpha$  and  $^3\text{He}$  clusters. The  $^7\text{Be}$  spin arises from the coupling of  $L$  with the total spin  $\frac{1}{2}$  of these clusters while the channel spin  $I$  is obtained after coupling with the proton spin. In configuration  $\alpha = 1$ , the  $^7\text{Be}$  nucleus is described in the GCM framework with a single generator coordinate  $R_2 = 3.7$  fm, and in configuration  $\alpha = 2$ , we adopt  $R_2 = 1.0$  fm for the

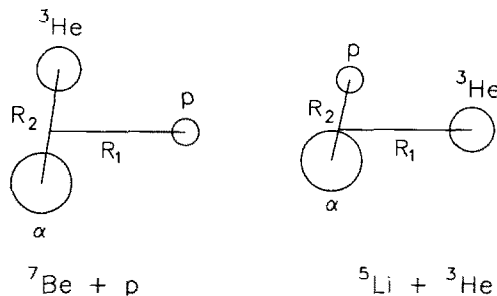


Fig. 1. Three-cluster configurations of  $^8\text{B}$ .

Table 1  
 ${}^7\text{Be}$  properties

| $m$                                                     | V1<br>0.6274 | V2<br>0.6351 | V3<br>0.6279 | V4<br>0.6247 | Exp. [28]       |
|---------------------------------------------------------|--------------|--------------|--------------|--------------|-----------------|
| $E(\frac{1}{2}^-)$ (MeV)                                | 0.43         | 0.43         | 0.43         | 0.43         | 0.43            |
| $E(\frac{7}{2}^-)$ (MeV)                                | 6.11         | 6.08         | 6.24         | 6.29         | 4.57            |
| $E(\frac{5}{2}^-)$ (MeV)                                | 6.95         | 6.91         | 7.09         | 7.13         | 6.73            |
| $Q(\frac{3}{2}^-)$ ( $e \cdot \text{fm}^2$ )            | -64          | -64          | -64          | -64          |                 |
| $\mu(\frac{3}{2}^-)$ ( $\mu_N$ )                        | -1.27        | -1.27        | -1.27        | -1.27        |                 |
| $B(E2, \frac{1}{2}^- \rightarrow \frac{3}{2}^-)$ (W.u.) | 52           | 52           | 52           | 52           |                 |
| $B(M1, \frac{1}{2}^- \rightarrow \frac{3}{2}^-)$ (W.u.) | 1.77         | 1.77         | 1.77         | 1.77         | $2.07 \pm 0.27$ |

$\alpha + p$  description of  ${}^5\text{Li}$ . We select for  $\alpha + {}^3\text{He}$  the orbital momenta  $L = 1$  and 3, which lead to the  $\frac{3}{2}^-$  ground state, and to the  $\frac{1}{2}^-$ ,  $\frac{7}{2}^-$  and  $\frac{5}{2}^-$  excited states of  ${}^7\text{Be}$ . For  ${}^5\text{Li}$ ,  $L = 1$  is chosen. The  ${}^7\text{Be} + p$  and  ${}^5\text{Li} + {}^3\text{He}$  relative motions are described in the GCM with generator coordinates  $R_1$  selected from 2.4 to 8.4 fm with a step of 1.2 fm. The numerical calculation of GCM matrix elements is explained in ref. [24].

The microscopic hamiltonian involves a central Volkov force [22], the exact Coulomb interaction, and a spin-orbit force [26] with strength  $S_0$ . The spin-orbit amplitude  $S_0$  is fitted to the experimental  $\frac{1}{2}^-$  excitation energy of  ${}^7\text{Be}$  or  ${}^7\text{Li}$ , yielding  $S_0 = 47.65 \text{ MeV} \cdot \text{fm}^5$  for  ${}^8\text{B}$  and  $S_0 = 53.07 \text{ MeV} \cdot \text{fm}^5$  for  ${}^8\text{Li}$ . In order to test the sensitivity of the results with respect to the nucleon-nucleon interaction, we consider the V1, V2, V3 and V4 forces simultaneously. For each Volkov force, we adjust the Majorana parameter  $m$  in order to reproduce the experimental  ${}^8\text{B}$  or  ${}^8\text{Li}$  binding energies with respect to the  ${}^7\text{Be} + p$  or  ${}^7\text{Li} + n$  thresholds. The Majorana parameters of the different Volkov forces are very similar. They are displayed in Tables 1 and 2 together with spectroscopic properties of  ${}^7\text{Be}$  and  ${}^7\text{Li}$ ,

Table 2  
 ${}^7\text{Li}$  properties

| $m$                                                     | V1<br>0.6234 | V2<br>0.6300 | V3<br>0.6265 | V4<br>0.6242 | Exp. [28]       |
|---------------------------------------------------------|--------------|--------------|--------------|--------------|-----------------|
| $E(\frac{1}{2}^-)$ (MeV)                                | 0.48         | 0.48         | 0.48         | 0.48         | 0.48            |
| $E(\frac{7}{2}^-)$ (MeV)                                | 6.12         | 6.08         | 6.24         | 6.29         | 4.63            |
| $E(\frac{5}{2}^-)$ (MeV)                                | 7.06         | 7.02         | 7.19         | 7.23         | 6.68            |
| $Q(\frac{3}{2}^-)$ ( $e \cdot \text{fm}^2$ )            | -40          | -40          | -40          | -40          | $-34 \pm 6$     |
| $\mu(\frac{3}{2}^-)$ ( $\mu_N$ )                        | 3.15         | 3.15         | 3.15         | 3.15         | 3.26            |
| $B(E2, \frac{1}{2}^- \rightarrow \frac{3}{2}^-)$ (W.u.) | 20           | 20           | 20           | 20           | $17 \pm 10$     |
| $B(E2, \frac{7}{2}^- \rightarrow \frac{3}{2}^-)$ (W.u.) | 13.4         | 13.4         | 13.4         | 13.4         | 4.3             |
| $B(M1, \frac{1}{2}^- \rightarrow \frac{3}{2}^-)$ (W.u.) | 2.41         | 2.41         | 2.41         | 2.41         | $2.75 \pm 0.14$ |

respectively. Electromagnetic properties of  ${}^7\text{Be}$  and  ${}^7\text{Li}$  do not depend on the interaction when a single generator coordinate is employed. They are in reasonable agreement with available experimental data, except for the  $B(E2, \frac{7}{2}^- \rightarrow \frac{3}{2}^-)$  value in  ${}^7\text{Li}$  which is overestimated by a factor of three.

### 3. The ${}^8\text{Li}$ nucleus and the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ reaction

#### 3.1. Spectroscopic properties of ${}^8\text{Li}$

Before discussing results obtained with the total wave functions (1), we first consider the energy of the  ${}^8\text{Li}$  nucleus as a function of the generator coordinates  $R_1$  and  $R_2$ . In other words, we select a single set of  $(R_1, R_2)$  values in Eq. (1), and compute the expectation value of the hamiltonian with the basis functions  $\Phi_{lll}^{JM\pi}$  corresponding to the different values of  $l$ ,  $L$  and  $I$  compatible with given  $J$  and  $\pi$ . The  ${}^5\text{He} + t$  configuration is neglected in this preliminary study. We present in Fig. 2 this energy surface for the  $2^+$  partial wave. Fig. 3 illustrates the  $0^+$ ,  $1^+$ ,  $3^+$  and  $4^+$  partial waves. In the ground-state partial wave, the  $R_2$  value of the minimum is larger than  $R_1$ . Even if this picture is schematic only, this means that  ${}^8\text{Li}$  should not be considered as a  ${}^7\text{Li}$  core with an external neutron. From Fig. 3, we deduce that the  $1^+$  structure resembles the  $2^+$  one, suggesting that the space dependences of the  $2^+$  and  $1^+$  wave functions should be very similar. In the  $0^+$  partial wave, the minimum is close to  $R_1 = R_2 = 4$  fm which means that a possible  $0^+$  resonance should be strongly clustered. The structure of the  $3^+$  and  $4^+$  states deserves a special attention. The  $3^+$  and  $4^+$  experimental states are not reproduced in a single-channel  ${}^7\text{Be}(L=1) + p$  model [9] but, according to the present calculation, they should have a  ${}^7\text{Be}(L=3) + p$  structure, in qualitative agreement

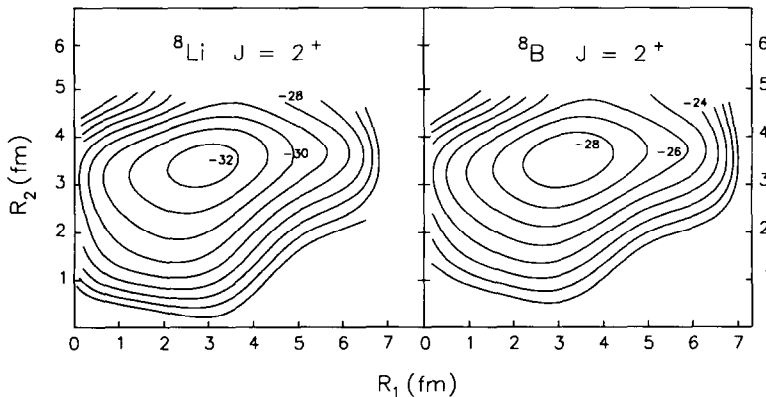


Fig. 2. Expectation energy of the  ${}^8\text{Li}$  and  ${}^8\text{B}$  ground states as a function of the  $R_1$  and  $R_2$  generator coordinates (see Fig. 1) for interaction V2. The contour lines are plotted by steps of 1 MeV.

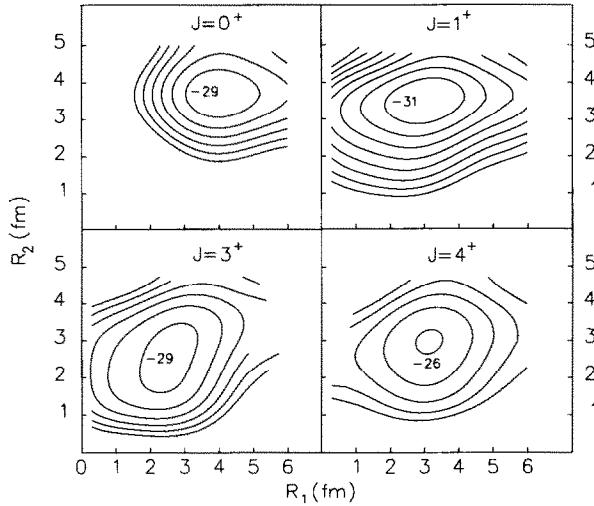


Fig. 3. Expectation energy of the  $0^+$ ,  $1^+$ ,  $3^+$  and  $4^+$  partial waves in  $^8\text{Li}$  as a function of the  $R_1$  and  $R_2$  generator coordinates (see Fig. 1) for interaction V2. The contour lines are plotted by steps of 1 MeV.

with ref. [21]. At the minimum of the binding energy, the  $R_1$  and  $R_2$  values are smaller than for the  $2^+$  ground state. This is an indirect indication for a small reduced width of the  $3^+$  and  $4^+$  excited states.

Now, we come back to the full GCM calculation. In Fig. 4, we present the energy spectra obtained with the wave functions (1) and with the V2 interaction.

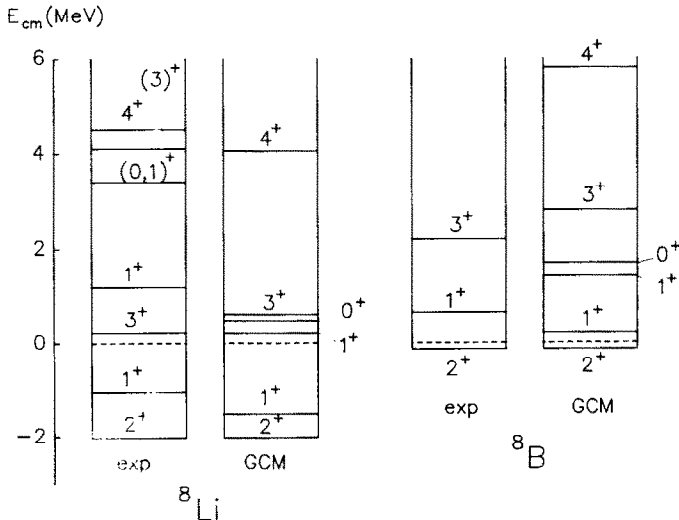


Fig. 4. GCM and experimental  $^8\text{Li}$  and  $^8\text{B}$  spectra obtained with the V2 interaction. The  $^7\text{Li}+n$  and  $^7\text{Be}+p$  thresholds are represented by dashed lines.

Table 3  
Spectroscopic properties of  $^8\text{Li}$

|                                                   | V1   | V2   | V3   | V4   | Exp. [28]                      |
|---------------------------------------------------|------|------|------|------|--------------------------------|
| $Q(2^+) (e \cdot \text{fm}^2)$                    | 3.1  | 3.1  | 3.1  | 3.1  | $2.4 \pm 0.2, 3.27 \pm 0.06^b$ |
| $\mu(2^+) (\mu_N)$                                | 1.41 | 1.39 | 1.43 | 1.45 | 1.65                           |
| $\Gamma_n(3^+)^a (\text{keV})$                    | 41   | 44   | 43   | 42   | $33 \pm 6$                     |
| $\Gamma_n(4^+)^a (\text{keV})$                    | 10   | 10   | 12   | 12   | $35 \pm 15$                    |
| $B(\text{M1}, 1^+ \rightarrow 2^+) (\text{W.u.})$ | 3.9  | 3.9  | 3.8  | 3.8  | $2.8 \pm 0.9$                  |
| $B(\text{M1}, 3^+ \rightarrow 2^+) (\text{W.u.})$ | 0.20 | 0.19 | 0.21 | 0.22 | $0.29 \pm 0.13$                |
| $B(\text{E2}, 1^+ \rightarrow 2^+) (\text{W.u.})$ | 3.7  | 3.6  | 3.7  | 3.7  | $47 \pm 23^c, 87 \pm 23^d$     |
| $B(\text{E2}, 3^+ \rightarrow 2^+) (\text{W.u.})$ | 9.2  | 9.3  | 9.1  | 9.1  |                                |
| $B(\text{E2}, 4^+ \rightarrow 2^+) (\text{W.u.})$ | 5.7  | 5.8  | 5.8  | 5.9  |                                |

<sup>a</sup> With energies fitted to experiment.

<sup>b</sup> From ref. [29].

<sup>c</sup> From ref. [17].

<sup>d</sup> From ref. [18].

Let us recall that only the binding energy of the ground state is fitted by the parameter  $m$  of the interaction. The excitation energy of the  $1^+$  state is somewhat underestimated. There are two possible reasons for this result: (i) the tensor force which is missing here, might affect the  $2^+ - 1^+$  energy difference, and (ii) our choice of the  $\alpha + t$   $R_2$  generator coordinate is intended to minimize the  $^7\text{Li}$  binding energy, rather than the  $^8\text{Li}$  binding energy. A careful examination of Figs. 2 and 3 suggests that the  $1^+$  excitation energy might be increased by taking a smaller  $R_2$  value. However, our main goal here is the description of capture reactions, and since the importance of computer times restricts us to a single  $R_2$  value, we decided to optimize the  $^7\text{Li}$  and  $^7\text{Be}$  wave functions. The second  $1^+$  resonance is also too low. As suggested by Fujiwara and Tang [21], the model nicely reproduces the  $3^+$  and  $4^+$  experimental energies. The inclusion of the spin-orbit force allows here a meaningful comparison with experiment. The model predicts a  $0^+$  resonance near 1 MeV, in good agreement with the shell-model calculation of Knox et al. [27]. It also suggests broad ( $\Gamma_n > 1$  MeV)  $1^-$  and  $2^-$  resonances (not presented in Fig. 4) near 1 MeV.

In Table 3, we give some spectroscopic properties of  $^8\text{Li}$ . For all quantities, the sensitivity with respect to the nucleon-nucleon interaction is weak. The quadrupole moment slightly overestimates the experimental value of ref. [28], but is in good agreement with a recent measurement of Minamisono et al. [29]. The magnetic moment is close to experiment, and improved with respect to our earlier calculation [9] where we found  $1.18 \mu_N$  with the V2 force. The  $3^+$  and  $4^+$  neutron widths have the correct order of magnitude. This is satisfactory if one keeps in mind that the  $^7\text{Li}(\frac{3}{2}^-) + n$  configuration is a small component of the total wave functions. Notice that the  $\Gamma_n$  values given in Table 3 are obtained with a fit of the Majorana parameters to the experimental energies. The accuracy of the M1 transition

probabilities supports the validity of the GCM  $1^+$ ,  $2^+$  and  $3^+$  wave functions. Therefore, the recent experimental value [17,18] for the E2 transition probability between the  $1^+$  and  $2^+$  states is difficult to understand. This problem is discussed in detail in ref. [19].

### 3.2. The ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ reaction

The  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$  reaction is important in non-homogeneous big-bang models but it also provides a valuable test of models dedicated to the mirror  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  reaction. There are currently two sets of data covering a wide energy range. The first measurement has been performed by Imhof et al. [14] for  $E_{\text{lab}} = 30$  to 1000 keV. Recently, Wiescher et al. [15] remeasured the cross section from 25 to 420 keV and found a reduction by a factor of two with respect to ref. [14]. However, their data are not consistent with the recent measurement of Nagai et al. [16] and with the  $1/v$  extrapolation of the thermal cross section [28].

The GCM results are presented in Fig. 5. For the V2 force, we find a cross section very similar to that obtained in the  ${}^7\text{Li} (L = 1) + n$  model [9], except near the  $3^+$  resonance, which was missing in that approach (note that here the Majorana parameter of the  $3^+$  partial wave is slightly modified in order to fit the resonance energy). This result shows that the non-resonant low-energy cross section does not depend much on the added configurations. There is a small dependence on the nucleon–nucleon interaction below the  $3^+$  resonance, but all our forces support the normalization of the Imhof et al. data. At the astrophysical

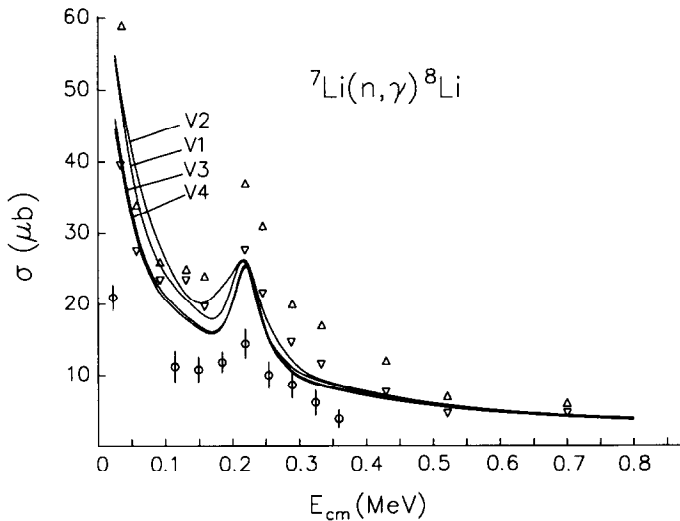


Fig. 5.  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$  cross section as a function of the c.m. energy, for the different Volkov forces. Triangles represent the data of Imhof et al. [14] (with two different normalizations), and circles represent the data of Wiescher et al. [15].

Table 4  
Spectroscopic properties of  $^8\text{B}$

|                                            | V1   | V2   | V3   | V4   | Exp. [28]         |
|--------------------------------------------|------|------|------|------|-------------------|
| $Q(2^+)$ ( $e \cdot \text{fm}^2$ )         | 7.2  | 7.3  | 7.1  | 7.0  | $6.83 \pm 0.21^b$ |
| $\mu(2^+)$ ( $\mu_N$ )                     | 1.42 | 1.45 | 1.39 | 1.38 | 1.03              |
| $\Gamma_p(1^+)^a$ (keV)                    | 58   | 59   | 55   | 54   | $37 \pm 5$        |
| $\Gamma_p(3^+)^a$ (keV)                    | 501  | 529  | 505  | 498  | $350 \pm 40$      |
| $B(\text{M}1, 1^+ \rightarrow 2^+)$ (W.u.) | 3.4  | 3.5  | 3.4  | 3.4  | $5.1 \pm 2.5$     |
| $B(\text{M}1, 3^+ \rightarrow 2^+)$ (W.u.) | 0.14 | 0.16 | 0.18 | 0.19 |                   |
| $B(\text{E}2, 1^+ \rightarrow 2^+)$ (W.u.) | 13.7 | 13.7 | 13.5 | 13.6 |                   |
| $B(\text{E}2, 3^+ \rightarrow 2^+)$ (W.u.) | 8.6  | 7.8  | 8.6  | 9.0  |                   |

<sup>a</sup> With energies fitted to experiment.

<sup>b</sup> From ref. [29].

energy  $E_{\text{c.m.}} = 25$  keV, we find cross sections equal to 54.2, 54.9, 46.0 and 44.5  $\mu\text{b}$  for the V1, V2, V3 and V4 interactions, respectively. The model supports the lower normalization of the Imhof et al. data (50  $\mu\text{b}$ ) and Nagai et al. data ( $39.3 \pm 6.0$   $\mu\text{b}$ ) rather than the Wiescher et al. data ( $19.6 \pm 1.8$   $\mu\text{b}$  deduced from  $E_{\text{lab}} = 25$  keV by a  $1/v$  law). These experiments seem to favour the V3 and V4 forces, but, because of some uncertainties of the model (choice of the  $R_2$  generator coordinate, lack of tensor force) we do not believe that the differences between the interactions are really significant.

#### 4. The $^8\text{B}$ nucleus and the $^7\text{Be}(p, \gamma)^8\text{B}$ reaction

##### 4.1. Spectroscopic properties of $^8\text{B}$

The  $^8\text{B}$  binding energy as a function of  $R_1$  and  $R_2$  is compared to the  $^8\text{Li}$  binding energy in Fig. 2. The shapes of the energy surfaces are essentially identical, but the minimum in  $^8\text{B}$  is slightly pushed up to larger  $R_1$  and  $R_2$  values. The Coulomb shift is about 4 MeV between  $^8\text{Li}$  and  $^8\text{B}$ . Fig. 4 displays the GCM  $^8\text{B}$  spectrum about which the comments are similar to those of  $^8\text{Li}$ . We predict  $0^+$  and  $1^+$  resonances, and a narrow ( $\Gamma_p \approx 20$  keV)  $4^+$  resonance near 6 MeV. In Table 4, we gather some spectroscopic properties of  $^8\text{B}$ . The results are weakly sensitive to the nucleon–nucleon interaction. As for  $^8\text{Li}$ , the theoretical quadrupole moment of the ground state nicely agrees with the recent data of Minamisono et al. [29]. These authors suggest that the large quadrupole moment of  $^8\text{B}$  provides evidence for the existence of a proton halo in this nucleus. They show that it can be explained from a long-range proton density, but with rather large effective charges. A parallel between this interpretation and the present calculation is not straightforward. The small proton-separation energy obviously causes a slow decrease of the proton wave functions. However, as for  $^8\text{Li}$ , Fig. 2 indicates that in a

${}^7\text{Be} + p$  description of  ${}^8\text{B}$ , the  ${}^7\text{Be}$  core is strongly clustered and that the additional proton remains close to this core, in contradiction with the halo picture. Without any effective charge, the present model explains simultaneously the  ${}^8\text{Li}$  and  ${}^8\text{B}$  quadrupole moments with rather similar wave functions. The larger  $Q(2^+)$  of  ${}^8\text{B}$  naturally arises from a larger number of protons and a slightly broader extension of the wave function due to Coulomb repulsion. Therefore, the notion of halo should be applicable to both nuclei or should not be applicable at all. In our opinion, the undeniable slow exponential decrease of the proton wave function in  ${}^8\text{B}$  does not really constitute a halo, in the sense of the neutron halos in  ${}^6\text{He}$  and  ${}^{11}\text{Li}$ .

The proton widths of the  $1^+$  and  $3^+$  resonances are slightly overestimated in the GCM, but the M1 transition probability  $B(\text{M1}, 1^+ \rightarrow 2^+)$  agrees with experiment within the error bars.

#### 4.2. The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ capture reaction

The determination of the  $S$ -factor at zero energy is a long-standing problem. In Fig. 5, we present the microscopic  $S$ -factor with the experimental data of refs. [30–32]. Following the prescription of Barker and Spear [3], these data are renormalized to a value  $\sigma_{\text{dp}} = 157$  mb for the  ${}^7\text{Li}(d, p){}^8\text{Li}$  cross section. For the  $1^+$  partial wave we have tuned the Majorana parameter in order to fit the experimental energy of the resonance. The  $S$ -factor contains the dominant E1 multipolarity and the M1 multipolarity which gives the resonant contribution near 0.6 MeV, due to the  $1^+$  state in  ${}^8\text{B}$ . According to the slight overestimation of the GCM width (see Table 4), the peak around this resonance is a little too broad. Beyond the  $1^+$  resonance, the GCM is consistent with the data which, however, significantly differ from one experiment to another one.

Let us now discuss the low-energy part of the  $S$ -factor. The energy dependence only depends on Coulomb functions, and is well known for many years [2]. The  $S(0)$  values are given in Table 5. The present  $S$ -factor obtained with the V2 force is identical to the value obtained in ref. [9]. As for  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ , there is therefore a stability of the  $S$ -factor with respect to the number of GCM basis wave functions. The different forces lead to  $S(0)$  values ranging from  $0.027 \text{ keV} \cdot \text{b}$  to  $0.030 \text{ keV} \cdot \text{b}$ . This means that there is at least a 10% uncertainty regarding the nucleon–nucleon

Table 5  
 ${}^7\text{Be}(p, \gamma){}^8\text{B}$   $S(0)$  values

| Interaction | $S(0)$ (keV · b) |
|-------------|------------------|
| V1          | 0.029            |
| V2          | 0.030            |
| V3          | 0.028            |
| V4          | 0.027            |

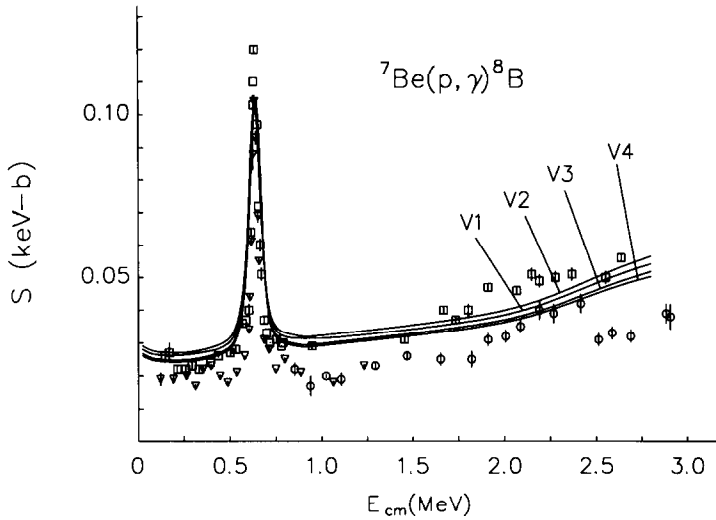


Fig. 6.  ${}^7\text{Be}(p, \gamma){}^8\text{B}$   $S$ -factors as a function of the c.m. energy. The data are taken from ref. [30] (squares), ref. [31] (circles) and ref. [32] (triangles), and normalized to  $\sigma_{\text{dp}} = 157$  mb.

interaction. This sensitivity is lower than predicted by Mukhamedzhanov and Timofeyuk [11] who investigate the asymptotic normalization of bound-state wave functions for different nuclei. They conclude that this quantity – and the cross section which is proportional to its square – significantly depends on the nucleon–nucleon force. However, as suggested in ref. [23], this sensitivity might decrease in the GCM when the binding energy of the clusters reaches, or is close to, saturation for the considered force. This proposal is supported by the present multiconfiguration calculation where the  $\alpha$  and  ${}^3\text{He}$  clusters are expected to be rather well reproduced by  $s$  orbitals with a common oscillator parameter equal to 1.5 fm. Of course, we only consider here Volkov forces involving gaussian terms, but the differences between the shapes of the four interactions seem large enough to allow some extrapolation to other forces. In ref. [23], squared normalization constants of asymptotic wave functions are investigated, with different forces, in the  $n + {}^{16}\text{O}$  weakly bound system. The V4 result is close to the mean value of the too large V2 result and experiment; it is very similar to that obtained with the HNY and Minnesota forces which are used by other authors [12]. This hierarchy seems to depend weakly on the considered system [33]. A transposition to the present case provides  $S(0) \geq 0.024$  keV · b. We therefore think that the  $S$ -factor at zero energy is not significantly different from the value used in astrophysics [1] ( $0.024 \pm 0.002$  keV · b), but that the small values proposed by Barker and Spear [3] (0.017 keV · b), and by Mukhamedzhanov and Timofeyuk [11] (0.0155 keV · b) are unlikely.

## 5. Conclusion

In the present paper, we have reconsidered the  ${}^8\text{Li}$  and  ${}^8\text{B}$  systems, by improving our earlier study [9] in different ways. We introduce here the  $L = 3$  states in the  ${}^7\text{Li}$  and  ${}^7\text{Be}$  nuclei, and take account of the  ${}^5\text{He} + t$  or  ${}^5\text{Li} + {}^3\text{He}$  configurations. In addition, we test the sensitivity of our results with respect to the choice of the nucleon–nucleon interaction. Of course, only Volkov forces are considered, but the differences between them are large enough to provide an estimate of the global accuracy. The GCM wave functions are shown to give a good description of most of the experimentally known  ${}^8\text{Li}$  and  ${}^8\text{B}$  states. In particular, the quadrupole moments of these mirror nuclei can be explained simultaneously without any clear indication of a proton halo in  ${}^8\text{B}$ . We find that spectroscopic properties are weakly dependent on the nucleon–nucleon interaction.

The  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$  and  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  non-resonant cross sections with the V2 force are very close to the previous ones [9] where only  ${}^7\text{Li}(L = 1) + n$  or  ${}^7\text{Be}(L = 1) + p$  configurations are taken into account. The present model reproduces the resonant part of the  ${}^7\text{Li}(n, \gamma){}^8\text{Li}$  cross section, due to the  $3^+$  resonance at  $E_{\text{c.m.}} = 0.22$  MeV. The GCM cross section supports the lower data normalization of Imhof et al. [14], rather than the normalization of Wiescher et al. [15]. The  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  cross section is more sensitive to the interaction choice but this dependence is weaker than predicted in ref. [11]. Therefore, the non-resonant  $S$ -factor should not be significantly different from the currently accepted value [1], so that smaller  $S(0)$  values [3,11] seem unlikely.

The present calculation might still be improved in two ways. The tensor force might partly remove the small inaccuracies in the GCM spectroscopic properties, such as the  $1^+$  excitation energies in  ${}^8\text{Li}$  and  ${}^8\text{B}$ . A second improvement would be to include more than one generator coordinate for the  ${}^7\text{Li}$  and  ${}^7\text{Be}$  wave functions. For computer time reasons, we are restricted to a single value, which we choose as minimizing the  ${}^7\text{Li}$  and  ${}^7\text{Be}$  binding energies. This description turns out to be realistic, in view of the spectroscopic properties of these nuclei. However, the present study suggests that in the ground state of  ${}^8\text{Li}$ , a smaller value might be more appropriate. Therefore, a simultaneous optimization of bound-state and scattering wave functions would require different (though rather close)  $R_2$  values. However, we do not believe that these extensions of the present model would bring qualitative changes to our conclusions.

We thank the Services de Programmation de la Politique Scientifique for a supercomputer grant on the CRAY-YMP computer of the ULB-VUB. One of us (P.D.) is grateful to Prof. H. Oberhummer for his hospitality in Vienna and for many helpful discussions concerning this work. We also thank N.K. Timofeyuk for

strongly emphasizing the importance of the force problem in the  ${}^7\text{Be}(p, \gamma){}^8\text{B}$  reaction.

## References

- [1] J.N. Bahcall and R.K. Ulrich, *Rev. Mod. Phys.* 60 (1988) 297
- [2] R.D. Williams and S.E. Koonin, *Phys. Rev. C* 23 (1981) 2773
- [3] F.C. Barker and R.H. Spear, *Astrophys. J.* 307 (1986) 847
- [4] T.A. Tombrello, *Nucl. Phys.* 71 (1965) 459
- [5] A. Aurdal, *Nucl. Phys.* A146 (1970) 385
- [6] R.G.H. Robertson, *Phys. Rev. C* 7 (1973) 543
- [7] F.C. Barker, *Aust. J. Phys.* 33 (1980) 177
- [8] K.H. Kim, M.H. Park and B.T. Kim, *Phys. Rev. C* 35 (1987) 363
- [9] P. Descouvemont and D. Baye, *Nucl. Phys.* A487 (1988) 420
- [10] E. Kolbe, K. Langanke and H.J. Assenbaum, *Phys. Lett. B* 214 (1988) 169
- [11] A.M. Mukhamedzhanov and N.K. Timofeyuk, *JETP Lett.* 51 (1990) 282
- [12] C.W. Johnson, E. Kolbe, S.E. Koonin and K. Langanke, *Astrophys. J.* 392 (1992) 320
- [13] R.A. Malaney and W.A. Fowler, *in* *Origin and distribution of the elements*, ed. G.J. Mathews (World Scientific, Singapore, 1988) p. 76
- [14] W.L. Imhof, R.G. Johnson, F.J. Vaughn and M. Walt, *Phys. Rev.* 114 (1959) 1037
- [15] M. Wiescher, R. Steininger and F. Käppler, *Astrophys. J.* A344 (1989) 464
- [16] Y. Nagai, M. Igashira, N. Mukai, T. Ohsaki, F. Uesawa, K. Takeda, T. Ando, H. Kitazawa, S. Kubono and T. Fukuda, *Astrophys. J.* A381 (1991) 444
- [17] R.J. Smith, J.J. Kolata, K. Lamkin, A. Morsad, F.D. Becchetti, J.A. Brown, W.Z. Liu, J.W. Jänecke, D.A. Roberts and R.E. Warner, *Phys. Rev. C* 43 (1991) 2346
- [18] J.A. Brown, F.D. Becchetti, J.W. Jänecke, K. Ashktorab, D.A. Roberts, J.J. Kolata, R.J. Smith, K. Lamkin and R.E. Warner, *Phys. Rev. Lett.* 66 (1991) 2452
- [19] P. Descouvemont and D. Baye, *Phys. Lett. B* 292 (1992) 235
- [20] Y. Fujiwara and Y.C. Tang, *Phys. Lett. B* 222 (1989) 311
- [21] Y. Fujiwara and Y.C. Tang, *Phys. Rev. C* 41 (1990) 28
- [22] A.B. Volkov, *Nucl. Phys.* 74 (1965) 33
- [23] D. Baye and N.K. Timofeyuk, *Phys. Lett. B* 293 (1992) 13
- [24] D. Baye and P. Descouvemont, *Proc. Int. Symp. on developments of nuclear cluster dynamics*, Sapporo, Japan, 1988, ed. Y. Akaishi et al. (World Scientific, Singapore, 1989) p. 6
- [25] P. Descouvemont and D. Baye, *Nucl. Phys.* A517 (1990) 143
- [26] D. Baye and N. Pecher, *Bull. Cl. Sc. Acad. Roy. Belg.* 67 (1981) 835
- [27] H.D. Knox, D.A. Resler and R.O. Lane, *Nucl. Phys.* A466 (1987) 245
- [28] F. Ajzenberg-Selove, *Nucl. Phys.* A490 (1988) 1
- [29] T. Minamisono, T. Ohtsubo, I. Minami, S. Fukuda, A. Kitagawa, M. Fukuda, K. Matsuta, Y. Nojiri, S. Takeda, H. Sagawa and H. Kitagawa, *Phys. Rev. Lett.* 69 (1992) 2058
- [30] R.W. Kavanagh, T.A. Tombrello, J.M. Mosher and D.R. Goosman, *Bull. Am. Phys. Soc.* 14 (1969) 1209
- [31] F.J. Vaughn, R.A. Chalmers, D. Kohler and L.F. Chase, *Phys. Rev. C* 2 (1970) 1657
- [32] B.W. Filippone, A.J. Elwyn, C.N. Davids and D.D. Koetke, *Phys. Rev. C* 28 (1983) 2222
- [33] L.D. Blokhintsev, A.M. Mukhamedzhanov, N.K. Timofeyuk and Yu.M. Tchuvilskii, *Izv. Akad. Nauk SSSR, ser. fiz.* 54 (1990) 569